

# Define Problem Statements

## Phase 1 · Brainstorming & Ideation

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This document defines the problem this capstone solves, written as user-centered problem-statement cards in the format "I am (user) ... I'm trying to ... but ... because ... which makes me feel ...". The cards cover the two primary users of the branch-based access control - the branch end-user (EEE User) and the ServiceNow administrator - plus the institution's data-security stakeholder.

### Problem Statement 1 - Branch end-user (EEE User)

|                     |                                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|
| I am                | A branch user (EEE) responsible for my department's student records in the Institution Details table.                              |
| I'm trying to       | View and manage only my own branch's records - create, read, update, and delete EEE students.                                      |
| But                 | Out of the box I can see and edit every branch's records - ECE, EEE, and CSE alike.                                                |
| Because             | Default table access is granted all-or-nothing by role, with no filter on the record's Branch value.                               |
| Which makes me feel | Exposed and anxious that I might read or change data I have no business touching, breaking students' privacy and record integrity. |

### Problem Statement 2 - ServiceNow administrator

|                     |                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| I am                | The ServiceNow administrator configuring access for the institution on the dev443713 instance.                                    |
| I'm trying to       | Enforce record-level access so each branch user reaches only the records whose Branch field matches their branch.                 |
| But                 | Standard role-based permissions cannot discriminate on a field value like Branch.                                                 |
| Because             | A plain role grants access to the whole u_institution_details table regardless of what each record contains.                      |
| Which makes me feel | That role-only security is insufficient and that I need field-value-driven ACLs, with security_admin elevation, to close the gap. |

### Problem Statement 3 - Institution data-security stakeholder

|               |                                                                                                |
|---------------|------------------------------------------------------------------------------------------------|
| I am          | The institution's data-security stakeholder accountable for how student data is exposed.       |
| I'm trying to | Guarantee that student records are visible and editable only to the authorized branch's staff. |

|                     |                                                                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------|
| But                 | Any user who can reach the table can currently read and edit all records.                             |
| Because             | Access is not scoped to the record's branch - there is no record-level, field-value control in place. |
| Which makes me feel | That we fall short of our data-privacy obligations until access is properly scoped per branch.        |